



THE UNIVERSITY OF
MELBOURNE

Marketing Research

Week 10





Recap

Types of Test	Purpose of Test
Chi-Square Analysis (X^2)	Test for relationship between two <u>nominal</u> variables.
Spearman Rank Order Correlation Coefficient	Test for relationship between two variables that are not nominal and that at least one is <u>ordinal</u> .
Pearson Product Moment Correlation Coefficient	Test for relationships between two <u>interval and/or ratio</u> variables.



Topics for today

- Bivariate Regression
- Multiple regression
- Assumptions for Regression Analysis
- Regression with dummy variables



Types of Regression Analysis

Types of Test	Purpose of Test
Bivariate Regression Analysis	Analyze the linear relationship between <u>an</u> independent variable and <u>an interval/ratio</u> dependent variable.
Multiple Regression Analysis	Analyze the linear relationship between <u>multiple</u> independent variables and <u>an interval/ratio</u> dependent variable.



Regression Analysis

Importance of regression analysis

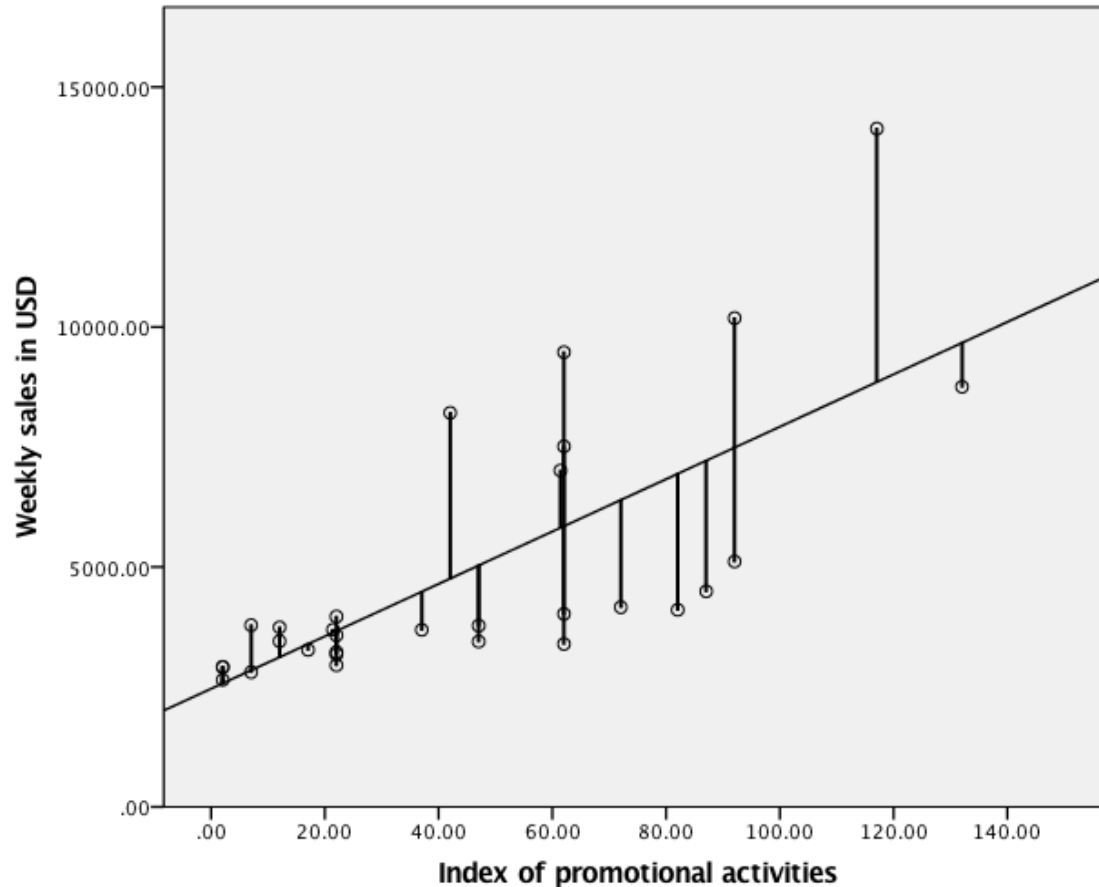
- Regression analysis is one of the most important methods in quantitative market research
- Marketing and strategy departments of major companies (e.g. , Procter & Gamble, BMW Group, Nestlé) use regression analysis regularly for decision-making

Regression analysis can

1. Indicate if the independent variables have a significant relationship with the dependent variable.
2. Indicate the relative strength of different independent variables' effects on a dependent variable.
3. Make predictions.

* In this lecture, we focus on linear regression (OLS)

The Regression Analysis Estimate a Line Using the Observations



The regression estimation procedure attempts to fit a best possible line. This line is optimal if the squared distances to all observation points are minimized. Thereby:

1. Positive and negative deviances don't balance each other
2. Bigger deviances are weighted disproportionately high

If the assumptions of the regression are harmed, the line might not be the true line!

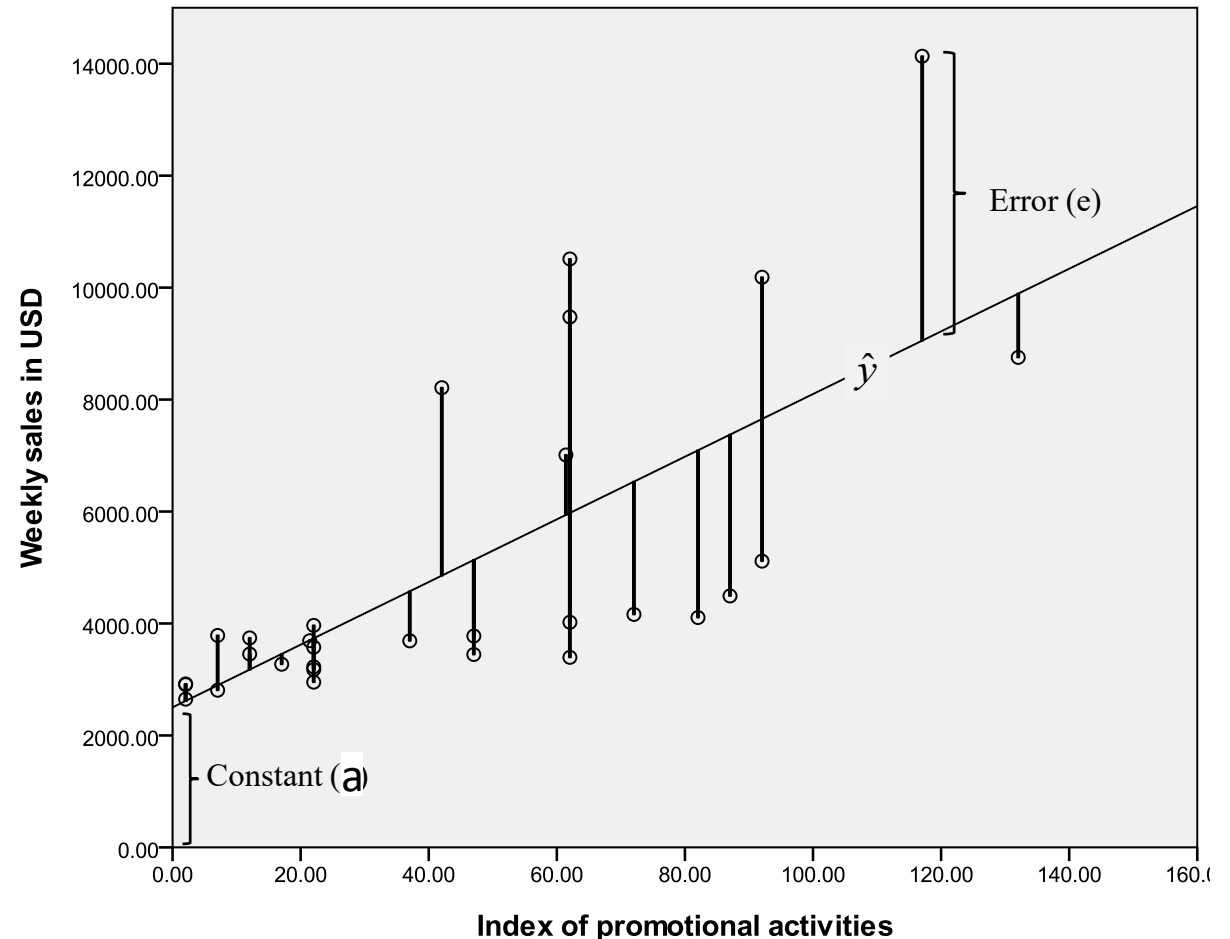
Understanding Regression Analysis

Notation

$$Y = a + b_1x_1 + e$$

Y	dependent variable
a	Intercept
b_1	Regression parameter (slope)
x_1	Independent variable
e	Residual

- When we use the symbol $\hat{}$, it indicates the predicted value. For instance, $\hat{y}$ means the predicted value of y
- Residual $e = y - \hat{y}$
- SPSS estimates a and b_1 based on data



Bivariate Regression Analysis

Example

70	s23	Numeric	8	0	... Oddjob Airways offers great value for money.
71	commitment	Numeric	8	2	Commitment

Value ▾	Label
1	Very unsatisfied
100	Very satisfied

Value ▾	Label
1	Fully disagree
7	Fully agree

- Two interval scaled variables → Bivariate Regression
 - Significance level: 5%
 - Independent variable: satisfaction with great value for money(s23)
 - Dependent variable: commitment (commitment)
 - Hypothesis: There is a linear association between commitment and satisfaction with the value.
- (*commitment is the average of 3 commitment variables (com1, com2, com3) measured on a 7-point Likert scale)



Bivariate Regression Analysis

Analyze → Regression → Linear → Select dependent variable (to dependent) and independent variable (to block)

The screenshot shows the SPSS 'Analyze' menu with 'Regression' selected, leading to 'Linear...'. The 'Linear Regression' dialog box is open, showing 'commitment' as the dependent variable and '... Oddjob Airways offers great value for money. [s23]' as the independent variable. A tip box indicates to right-click any variable and select 'Display variable names'.

Analyze → **Regression** → **Linear...**

Linear Regression

Dependent: commitment

Block 1 of 1

Block 1 of 1

... Oddjob Airways offers great value for money. [s23]

Method: Enter

Selection Variable: Rule...

Case Labels:

WLS Weight:

Reset Paste

*Tip: Right click any variable and select 'Display variable names'



Regression Outputs (in SPSS)

Variables Entered/Removed

- This table is for variable selection. You can ignore it for now.

Model Summary

- R-square
- Adjusted R-square (mainly for multiple regression)

ANOVA

- Sig. : p-value for the (Omnibus) F-test

Coefficients Table

- B (Unstandardized Coefficients) → coefficients in the regression equation
- Beta (Standardized Coefficients) → relative importance
- Sig. → p-value to test whether the individual slope coefficient is different from zero

Regression Output 1: Model Summary

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.459 ^a	.210	.209	1.54634

a. Predictors: (Constant), ... Oddjob Airways offers great value for money.

R-square (coefficient of determination, measuring goodness of fit)

- Range: from 0.00 to 1.00
- The percentage of variance of Y explained by X → the higher the more predictive the model
- 0.210 means “21.0 % of the variation in the commitment can be explained by satisfaction with the value for money (s23).”
- Recall that R = Pearson Correlation between Y (actual) and $\hat{Y}$ (predicted)

Regression Output 2: ANOVA

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	676.643	1	676.643	282.975	<.001 ^b
	Residual	2541.819	1063	2.391		
	Total	3218.463	1064			

a. Dependent Variable: Commitment

b. Predictors: (Constant), ... Oddjob Airways offers great value for money.

Null hypothesis (H_0): every slope coefficient is equal to zero (i.e., every $b = 0$)

Alternative hypothesis (H_1): at least one slope coefficient is not equal to zero

$p\text{-value} < 0.05$, so we reject H_0 .

- This means at least one regression slope coefficient is statistically different from zero. In this regression, there is only one X: rating, so the coefficient of “s23” is significantly different from zero.
- Knowing the value of X helps predict the value of Y to some degree (depending on the magnitude of R-square)

Regression Output 3: Coefficients Table

Coefficients^a

$H_0: b_1 = 0$

$H_1: b_1 \neq 0$

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.445	.113		21.717	<.001
	... Oddjob Airways offers great value for money.	.035	.002	.459	16.822	<.001

a. Dependent Variable: Commitment

- Use Coefficient B to formulate the regression equation:

$$\text{Commitment} = 2.445 + 0.035 * \text{Satisfaction for value for money (s23)}$$
- Interpretation: an increase of X by one unit is associated with an expected b units change in the outcome Y
- p-values (In column "Sig."): Each row is a one-sample t test against test value 0. If p-value $\leq \alpha$, then reject H_0 . This means that the regression coefficient in the same row is statistically different from zero.



Bivariate Regression Analysis

How to report the results of Bivariate Regression Analyses?

Example

In this test we hypothesized that satisfaction with the value for money affects commitment. The result of a Bivariate Regression Analysis shows that satisfaction with the value for money has a significant impact on commitment ($B = 0.035$, $p < .001$). Thus, the hypothesis is supported.



Multiple Regression

Multiple Regression Analysis

Example

42	s9	Numeric	8	0	... Oddjob Airways gives you a sense of safety.	{1, Very uns...
44	s10	Numeric	8	0	... the condition of Oddjob Airways's aircraft is immaculate.	{1, Very uns...
62	s19	Numeric	8	0	... Oddjob Airways pays attention to its service delivery's details.	{1, Very uns...
66	s21	Numeric	8	0	... Oddjob Airways makes traveling uncomplicated.	{1, Very uns...
70	s23	Numeric	8	0	... Oddjob Airways offers great value for money.	{1, Very uns...
71	commitm...	Numeric	8	2	Commitment	None

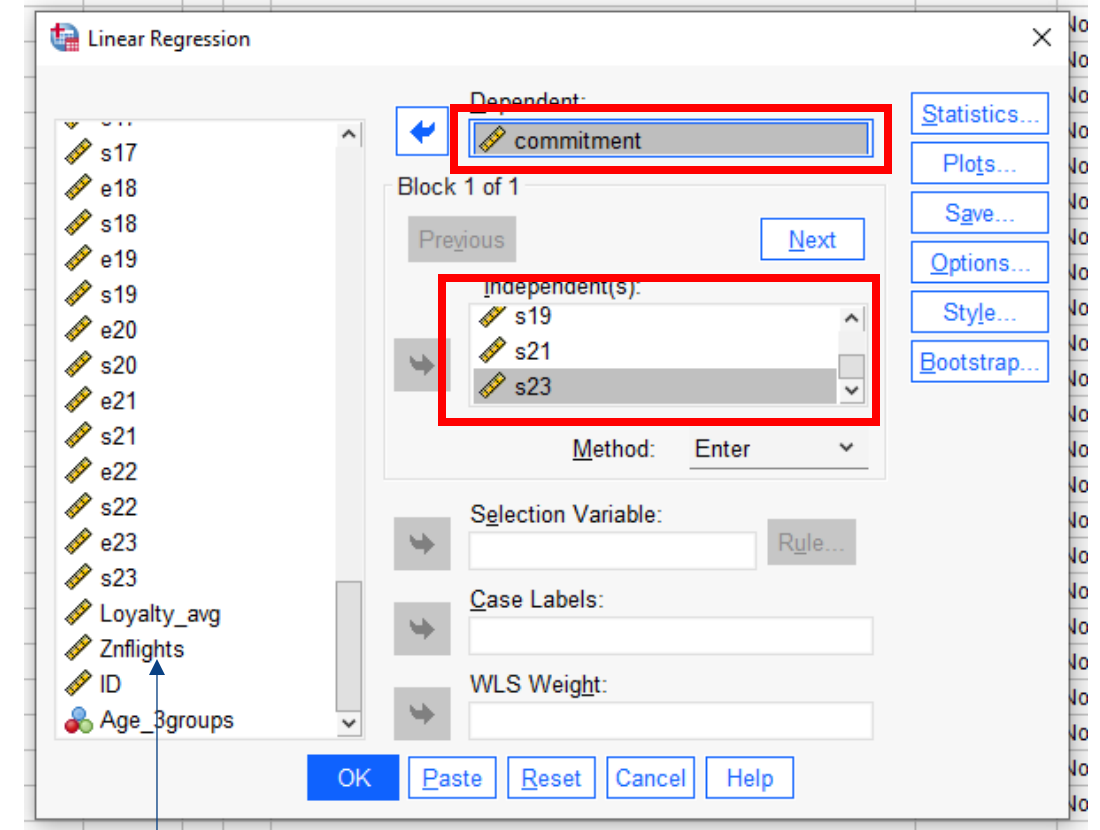
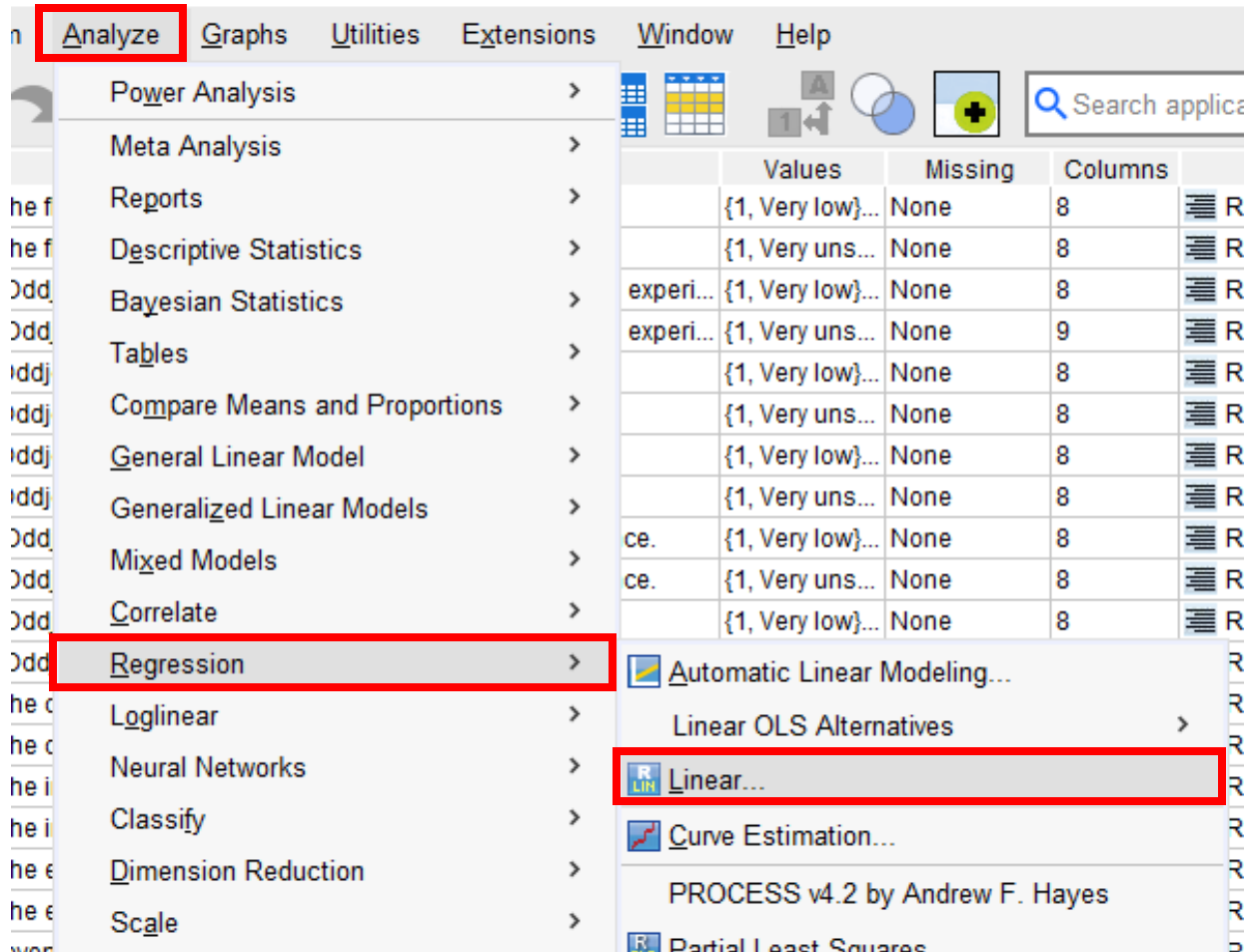
Independent
variables

Dependent
variables

- Multiple interval scale independent variables + Interval scale dependent variable → Multiple Regression
- Significance level: 5%
- Hypothesis: Customer satisfaction with the sense of safety, aircraft condition, attention to service delivery details, simplicity of traveling, and value for money affect their commitment to Oddjob Airways.

Multiple Regression Analysis

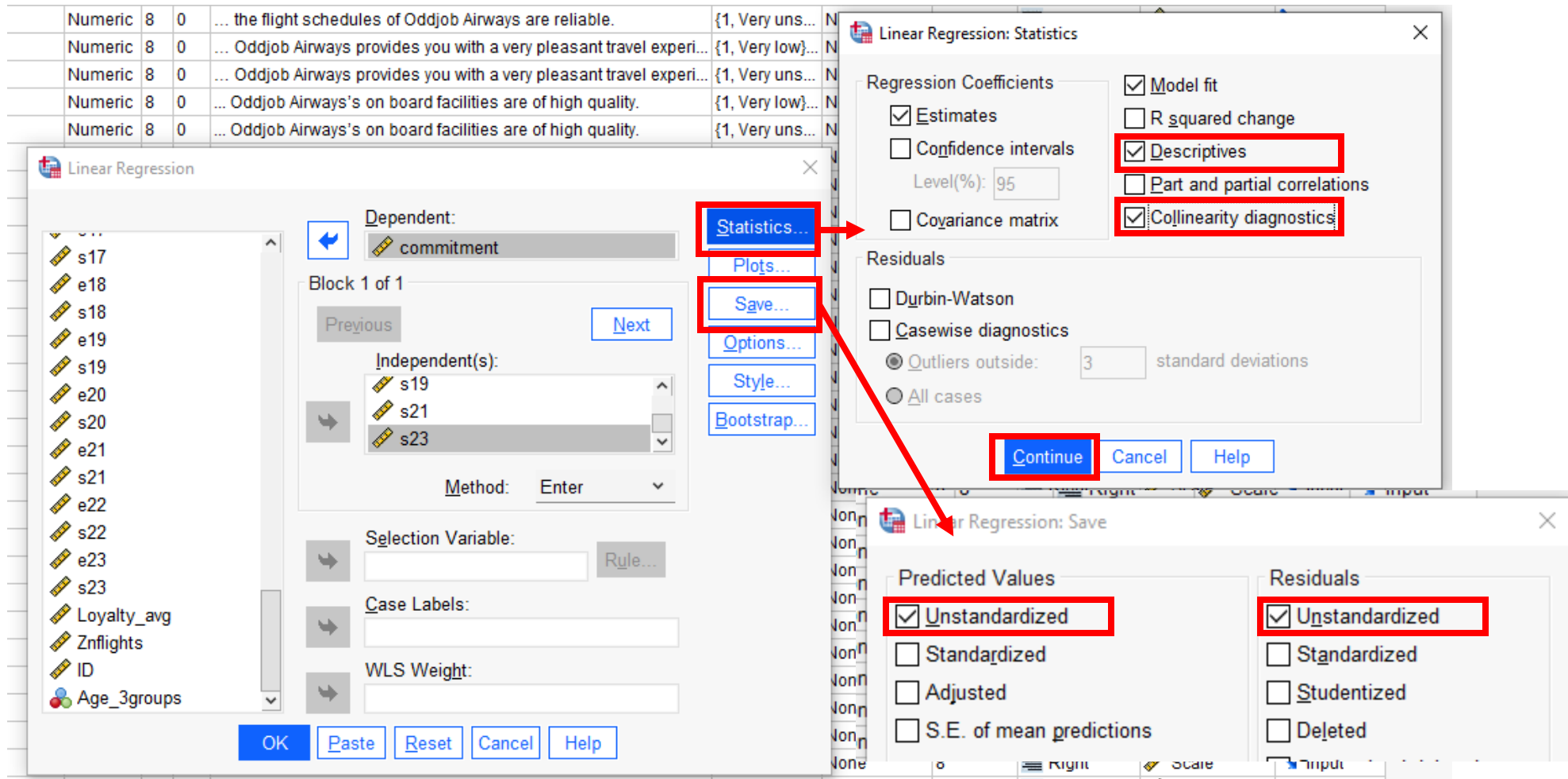
Analyze → Regression → Linear → Select dependent variable (to dependent) and independent variables (to block) (cont.)



*Tip
Right click any variable and
select 'Display variable names'

Multiple Regression Analysis

[Statistics] → Tick Descriptives and Collinearity diagnostics → continue → [Save] → Tick Unstandardized and Unstandardized



The image displays three overlapping SPSS dialog boxes for a Linear Regression analysis, with red boxes and arrows highlighting specific settings as instructed in the text above.

Linear Regression Dialog: The dependent variable is 'commitment'. Independent variables are 's19', 's21', and 's23'. The method is 'Enter'. The 'Statistics...' button is highlighted with a red box.

Linear Regression: Statistics Sub-dialog: This sub-dialog is open, showing options for regression coefficients and residuals. The 'Descriptives' and 'Collinearity diagnostics' checkboxes are highlighted with red boxes. The 'Continue' button is also highlighted with a red box.

Linear Regression: Save Sub-dialog: This sub-dialog is open, showing options for saving predicted values and residuals. The 'Unstandardized' checkboxes for both 'Predicted Values' and 'Residuals' are highlighted with red boxes.

Regression Output 1: Model Summary

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.566 ^a	.320	.317	1.41814

a. Predictors: (Constant), ... Oddjob Airways offers great value for money., ... Oddjob Airways gives you a sense of safety., ... Oddjob Airways makes traveling uncomplicated., ... the condition of Oddjob Airways's aircraft is immaculate., ... Oddjob Airways pays attention to its service delivery's details.

b. Dependent Variable: Commitment

32.0% of the variation in the DV is explained by the model (combination of s9, s10, s19, s21, and s23)

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.459 ^a	.210	.209	1.54634

a. Predictors: (Constant), ... Oddjob Airways offers great value for money.

In the bivariate regression, 21.0% of the variation in the DV is explained by the model (only s23)



Why do we need an Adjusted R^2 ?

Properties of R^2

- Measures the strength of association (or predictive power) in a multiple regression
- It usually increases as more independent variables are included

Adjusted- R^2

- Adjusts for the number of independent variables and sample size to account for diminishing returns → penalizing too many independent variables (overfitting)

In practice, we choose the best model based on both R^2 and Adjusted- R^2

Regression Output 2:

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	915.367	5	183.073	91.031	<.001 ^b
	Residual	1944.747	967	2.011		
	Total	2860.114	972			

a. Dependent Variable: Commitment

b. Predictors: (Constant), ... Oddjob Airways offers great value for money., ... Oddjob Airways gives you a sense of safety., ... Oddjob Airways makes traveling uncomplicated., ... the condition of Oddjob Airways's aircraft is immaculate., ... Oddjob Airways pays attention to its service delivery's details.

P-value $\leq .05$

The model is valid. This means at least one regression coefficient is statistically different from zero. (i.e., Model has predictive power, so the R-Square is significantly different from zero)

Regression Output 3: Coefficients Table

Coefficients^a

Model		Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics	
							Tolerance	VIF
1	(Constant)	1.164	.175		6.640	<.001		
	... Oddjob Airways gives you a sense of safety. s9	.006	.003	.068	1.859	.063	.522	1.916
	... the condition of Oddjob Airways's aircraft is immaculate. s10	.001	.003	.012	.324	.746	.504	1.983
	... Oddjob Airways pays attention to its service delivery's details. s19	.012	.003	.154	4.052	<.001	.487	2.051
	... Oddjob Airways makes traveling uncomplicated. s21	.019	.003	.250	6.878	<.001	.533	1.876
	... Oddjob Airways offers great value for money. s23	.015	.003	.203	5.887	<.001	.590	1.696

a. Dependent Variable: Commitment

P-value $\leq .05$

Attention to service delivery details (s19), simplicity of traveling (s21), and value for money (s23) affect commitment to Oddjob Airways

Regression Output 3: Coefficients Table

Coefficients^a

Model		Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance	VIF
1	(Constant)	1.164	.175		6.640	<.001		
	... Oddjob Airways gives you a sense of safety.	.006	.003	.068	1.859	.063	.522	1.916
	... the condition of Oddjob Airways's aircraft is immaculate.	.001	.003	.012	.324	.746	.504	1.983
	... Oddjob Airways pays attention to its service delivery's details. s19	.012	.003	.154	4.052	<.001	.487	2.051
	... Oddjob Airways makes traveling uncomplicated. s21	.019	.003	.250	6.878	<.001	.533	1.876
	... Oddjob Airways offers great value for money. s23	.015	.003	.203	5.887	<.001	.590	1.696

a. Dependent Variable: Commitment

- An increase of one unit in attention to service delivery details (s19) results in a commitment increase of 0.012, holding everything else constant.
- An increase of one unit in simplicity of traveling (s21) results in a commitment increase of 0.019 (holding everything else constant).
- An increase of one unit in value for money (s23) results in a commitment increase of 0.015 (holding everything else constant).

Regression Output 3: Coefficients Table

		Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	1.164	.175		6.640	<.001		
	... Oddjob Airways gives you a sense of safety.	.006	.003	.068	1.859	.063	.522	1.916
	... the condition of Oddjob Airways's aircraft is immaculate.	.001	.003	.012	.324	.746	.504	1.983
	... Oddjob Airways pays attention to its service delivery's details. s19	.012	.003	.154	4.052	<.001	.487	2.051
	... Oddjob Airways makes traveling uncomplicated. s21	.019	.003	.250	6.878	<.001	.533	1.876
	... Oddjob Airways offers great value for money. s23	.015	.003	.203	5.887	<.001	.590	1.696

a. Dependent Variable: Commitment

Among the three significant independent variables, simplicity of traveling (s21) has the biggest influence on commitment

The higher the absolute value of Beta (Standardized Coefficient), the more important it is to predict the dependent variable.



Collinearity/Multi-collinearity

Collinearity occurs when two or more predictors in a regression model are highly correlated.

Why It Matters

- Inflates standard errors → unstable coefficient estimates.
- Undermines the interpretability of individual predictor effects.

Detecting Collinearity

- Variance Inflation Factor (VIF) > 10 (or $VIF > 5$ for small sample) signals a problem.

Example

- Marketing spend and promotional discount rates in the retail industry often rise together, making it hard to isolate each effect on sales.

Mitigation Strategies

- Remove or combine correlated variables.
- Use principal component analysis (PCA) to create uncorrelated components.

Regression Output 3: Coefficients Table

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	1.164	.175		6.640	<.001		
	... Oddjob Airways gives you a sense of safety.	.006	.003	.068	1.859	.063	.522	1.916
	... the condition of Oddjob Airways's aircraft is immaculate.	.001	.003	.012	.324	.746	.504	1.983
	... Oddjob Airways pays attention to its service delivery's details.	.012	.003	.154	4.052	<.001	.487	2.051
	... Oddjob Airways makes traveling uncomplicated.	.019	.003	.250	6.878	<.001	.533	1.876
	... Oddjob Airways offers great value for money.	.015	.003	.203	5.887	<.001	.590	1.696

a. Dependent Variable: Commitment

Collinearity diagnostics show all VIFs lower than the threshold of 10.

→ *No collinearity is present*

** if there's any VIF higher than 10, navigate to the 'Correlations' table and identify highly correlated variables*



Multiple Regression Analysis

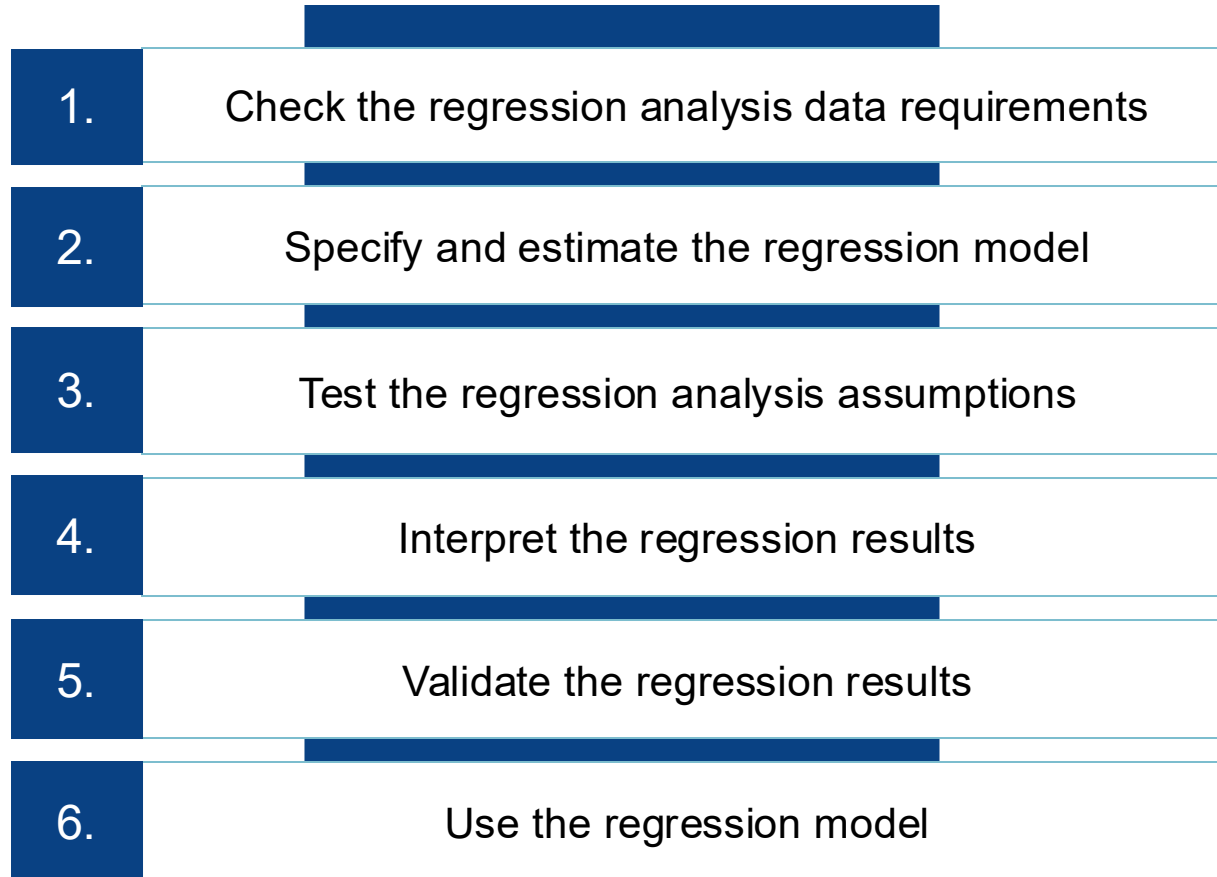
How to report the results of Multiple Regression Analyses?

Example

In this test, we hypothesized that customer satisfaction with the sense of safety, aircraft condition, attention to service delivery details, simplicity of traveling, and value for money affect their commitment to Oddjob Airways. The results of a multiple Regression Analysis show that satisfaction with attention to service delivery details ($B = 0.154$, $p < .001$), simplicity of traveling ($B = 0.250$, $p < .001$), and value for money ($B = 0.203$, $p < .001$) have a significant impact on commitment. Thus, the hypothesis is supported.



The Process of Conducting a Regression Analysis





1. Check the regression analysis data requirements

Sample size

- Formal power analysis (e.g. using G*power)

Enough variation in your data

Scale type of the dependent variable: interval or ratio scaled

Collinearity

- Collinearity is a data issue that arises if two independent variables are *highly correlated*.
- The VIF, or Variance Inflation Factor, can indicate collinearity. Values over 10 indicate collinearity.
- If collinearity is present, either use factor analysis, *re-specify the regression model*, or simply accept the collinearity (noting its presence).
 1. variable elimination: remove one of the highly correlated variable; or
 2. variable reduction: combine the highly correlated variables into one variable, or factor analysis



2. Specify and estimate the regression model

The goal is to build a simple yet complete model (e.g., SPSS).

Some suggestions:

- If independent variables have overlap, then focus on the most important or distinct variable.
- If you need to use a model for different circumstances, ensure the independent variables are the same to allow comparison.
- Consider the type of advice you want to give.
- Consider the sample sizes and rules of thumb. Smaller datasets require fewer independent variables
- If independent variables are nominal/ordinal, try to use **dummy variables**.



3. Test the regression analysis assumptions

Critical assumptions

1. The regression model can be expressed linearly,
2. The regression model's expected mean error is zero,
3. The errors' variance is constant (homoscedasticity), and
4. The errors are independent (e.g., no autocorrelation).

Optional assumption (to determine the significance of parameters)

5. The errors need to be approximately normally distributed.



3-1. Linearity

Linearity is an assumption that asks if we can write the regression model as

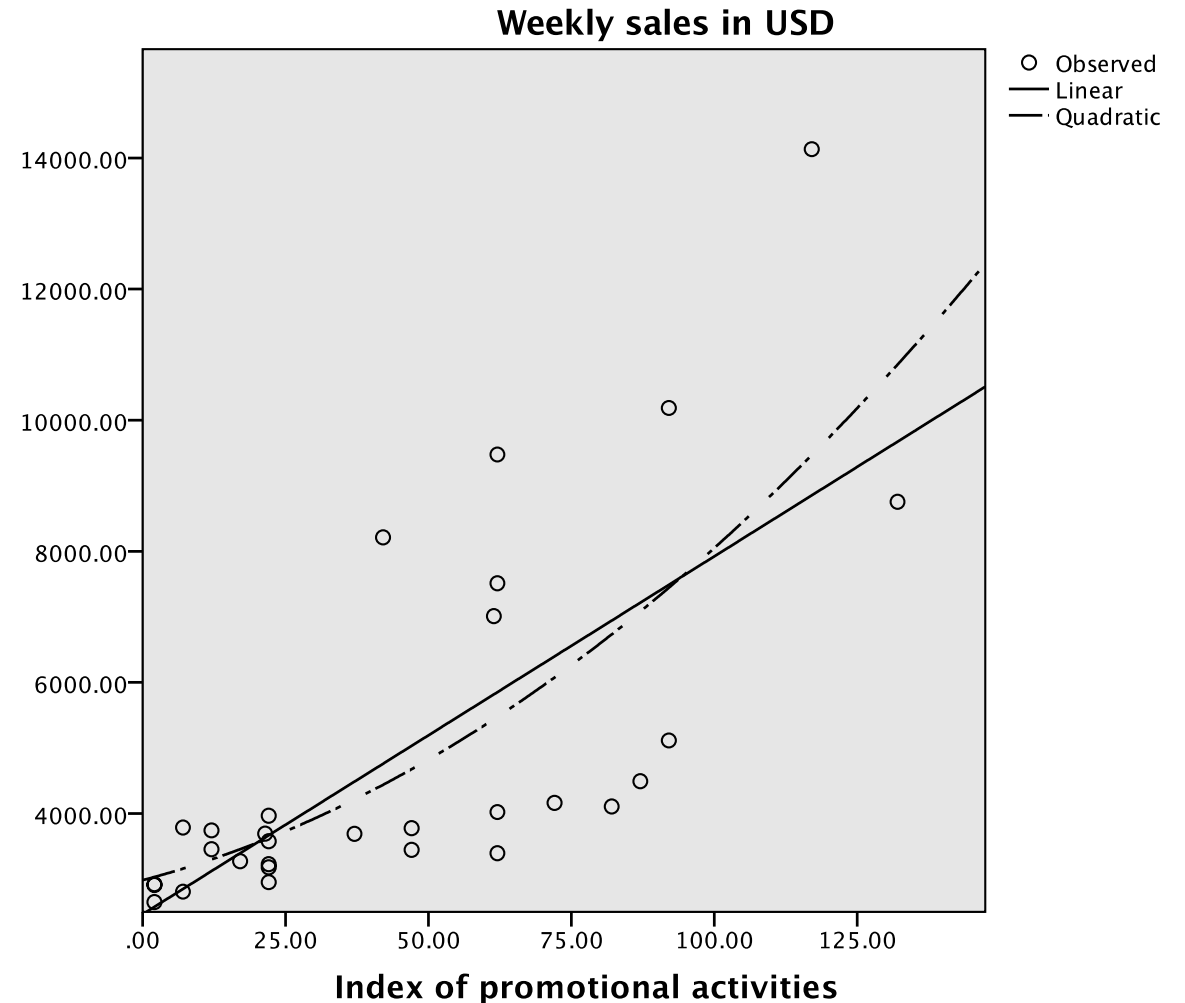
$$y = \alpha + \beta_1 x_1 + \dots + \beta_z x_z + e$$

A separate issue is if the relationship between Y (dependent variable) and X (independent variable) is best expressed as a linear relationship.

If not, we can use transformations:

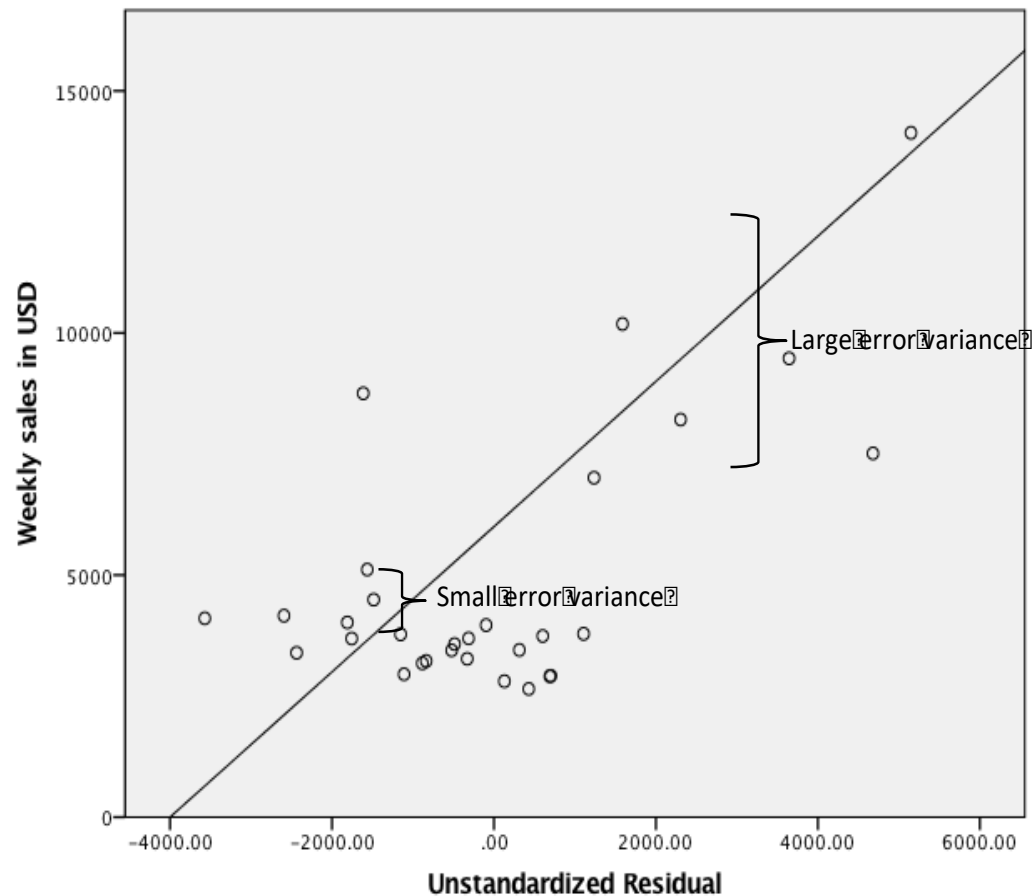
- x^2
- $\log(x)$

After such transformations, the model is still linear!

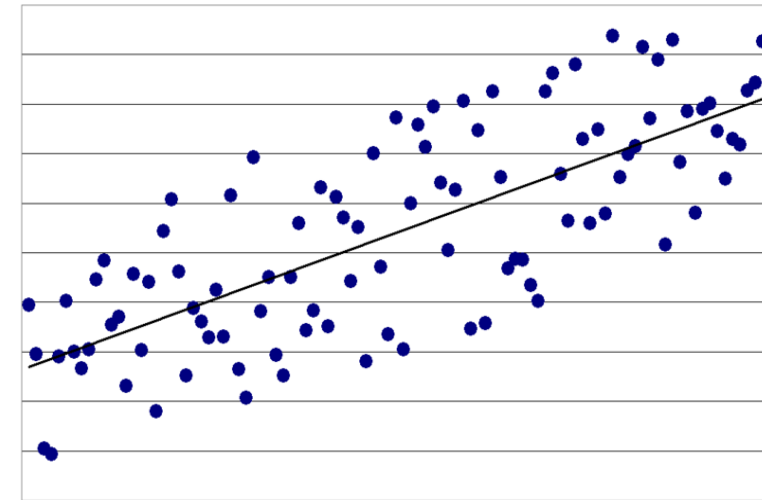


3-2. Homoskedasticity

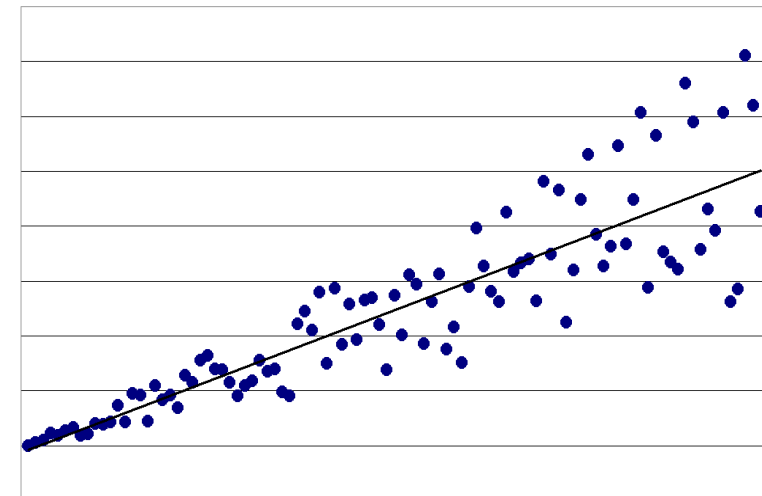
Is the variance constant or de(in)creasing as a function of one or more independent variable(s)?



Homoskedastic distribution



Heteroskedastic distribution





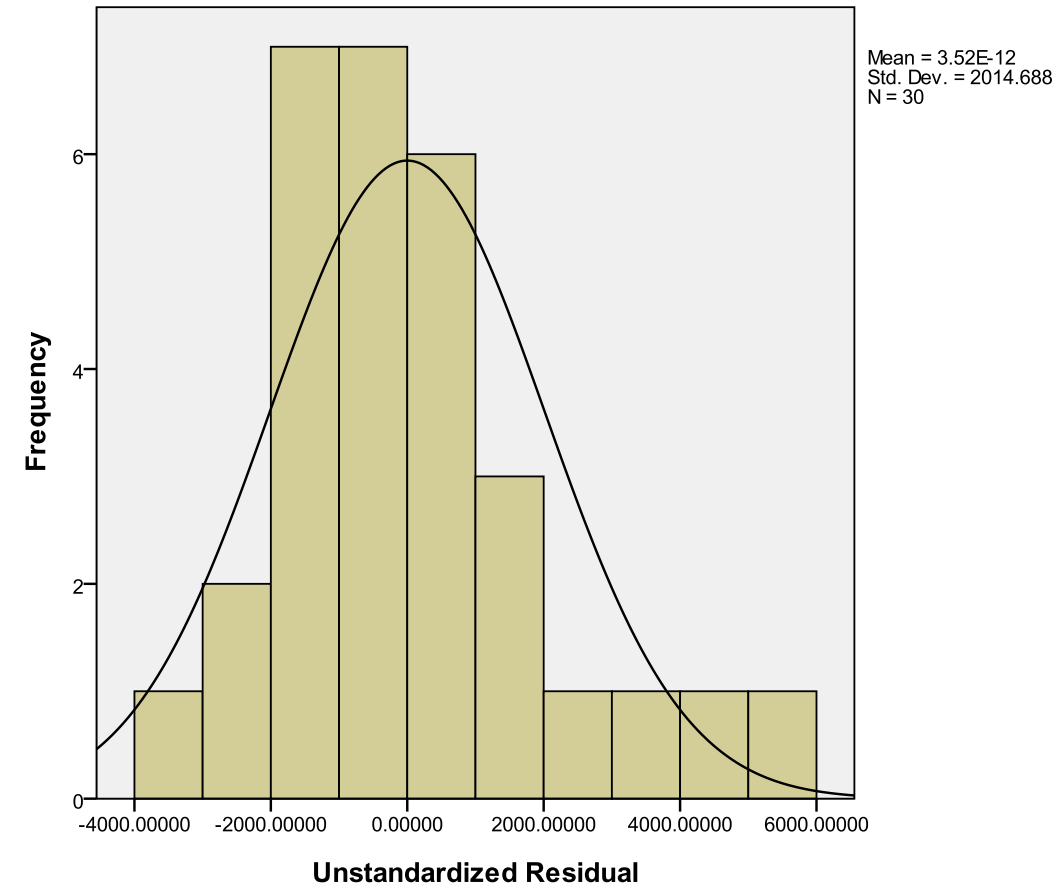
3-3. Are the Errors Independent?

- Example of correlated errors (or is there interdependence among the observations?)
 - Serial correlation. For example,
 - Monthly satisfaction of the same customer
 - Monthly sales of the same product
- There are tests and remedies for this type of analysis, but it is out of the scope of MR.



3-4. [Optional] Are the Residuals Approximately Normally Distributed?

- This optional assumption can be visualized using a histogram of the residuals.
- Formally, one can conduct a Shapiro-Wilk test on the residuals.
 - H_0 : The data are normally distributed
 - If the p-value > 0.05 , you fail to reject $H_0 \rightarrow$ assume normality; if the p-value ≤ 0.05 , you reject $H_0 \rightarrow$ data are not normally distributed



Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Unstandardized Residual	.131	30	.200 [*]	.939	30	.084

a. Lilliefors Significance Correction

*. This is a lower bound of the true significance.



4. Interpret the regression results

Effects of Individual Variables

1. The null and alternative hypotheses for each coefficient:

- $H_0: b_i = 0$
- $H_1: b_i \neq 0$
- If the *Sig.* column in SPSS indicates a value below .05, we consider this to be significant.

2. If the coefficient is significantly different from zero, report:

- The direction (+ / -)
- The magnitude of the coefficients and the relative importance.
 - Unstandardized (B): expresses the effect of a one-unit change in the independent variable on the dependent variable.
 - The standardized coefficient (Beta): the effect relative to the other variables. Standardized effects can be compared but not for dummy (binary) variables

Significance?
(p-value)

Direction?
(+ vs. -)

Size?
(b vs. standardized β)



5. Validate the regression model

Split-sample validation

- Randomly split the sample into 70% (**estimation sample**) and 30% (**validation sample**)
- Estimate the regression using the estimation sample and validate the model in the validation sample (e.g., see if the predicted values are close to the observed DV values).

Control variables

- If the regression outputs are stable, after you include additional independent variables (e.g., control variables)

6. Use the regression model

Interpretation of the regression model (regression equation) and prediction of dependent variables.

Example: based on the regression output below, what are the predicted sales if price = \$1.10 and the index of promotional activities = 50?

Coefficients ^a						
		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	29011.585	18448.456		1.573	.127
	Average supermarket price	-24003.037	16694.676	-.241	-1.438	.162
	Index of promotional activities	44.227	13.567	.547	3.260	.003

a. Dependent Variable: Weekly sales in USD

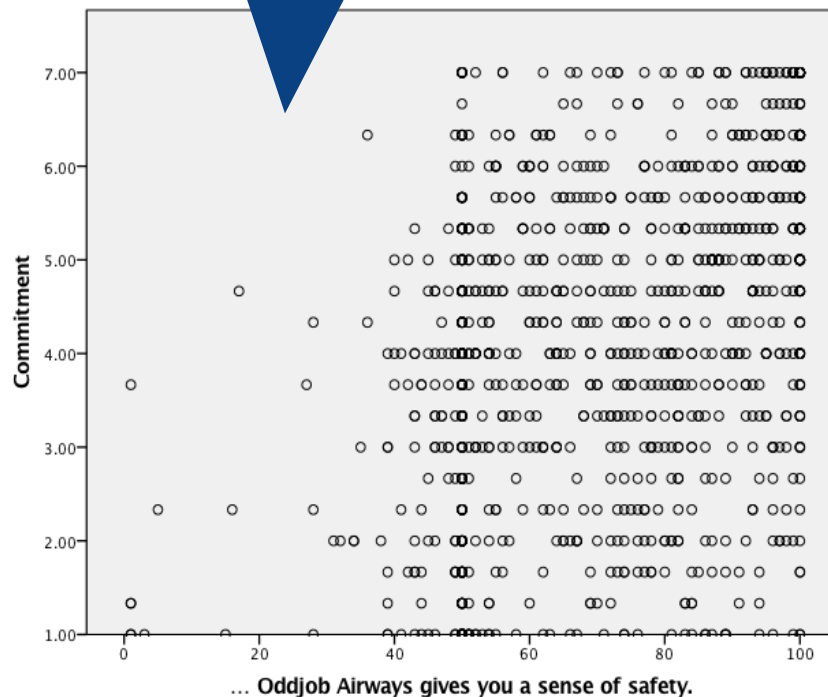
Regression model: Weekly Sales = 29011.585 – 24003.037*Avg Price + 44.227*Index of Promotional Activities
 → Predicted sales = 29,011.585 – 24,003.037 · **1.10 USD** + 44.227 · **50 promotional activities** = 4,819.594 USD sales per week

Visualize the data

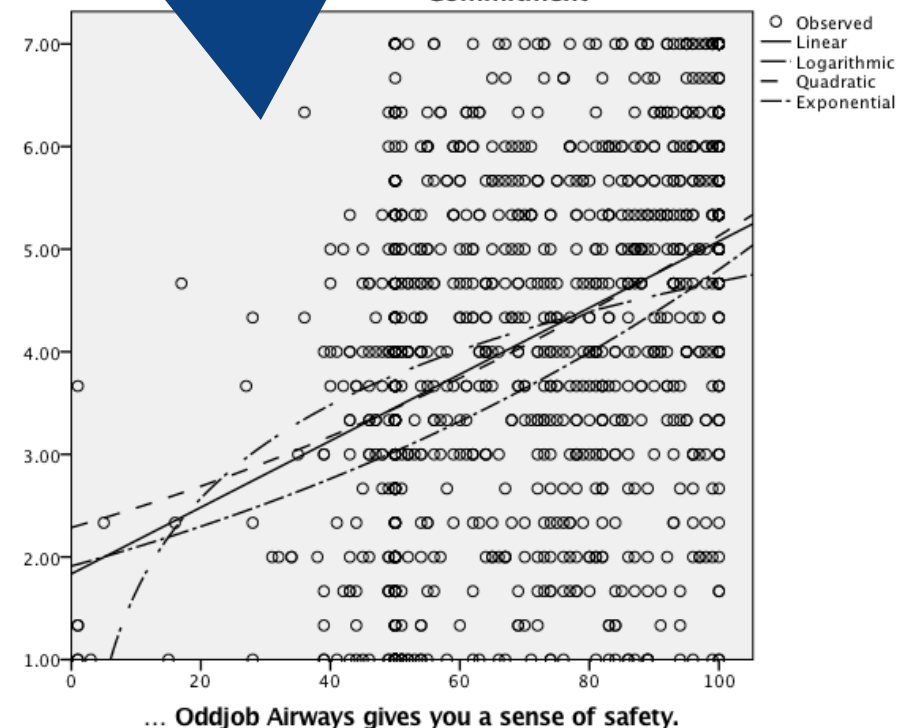
Create scatterplots of the independent variables against the dependent variable.

This is to visualize the data instead of testing the linearity assumption.

No clear conclusion,
which is typical



Add the trend lines



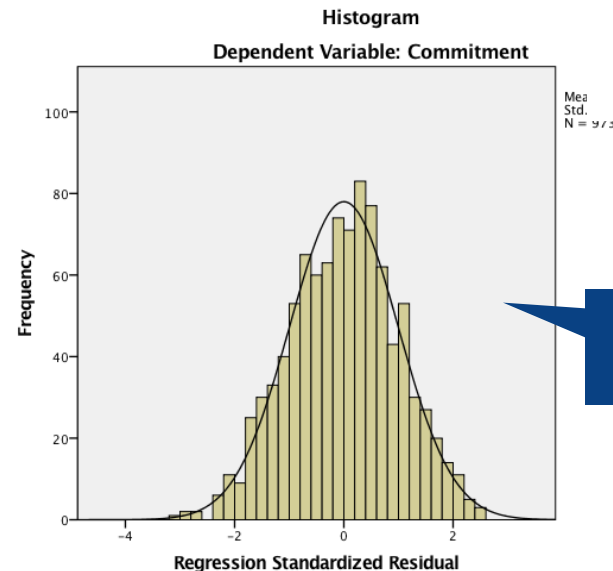
Visualize the residuals (Assumptions)

Homoskedasticity

The variance of the errors is constant

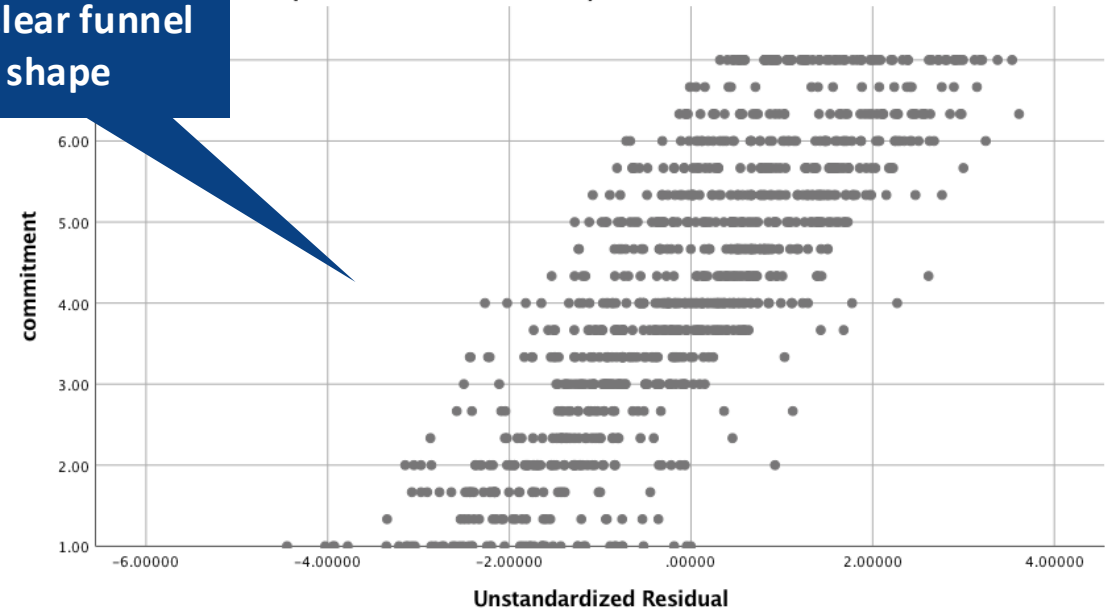
- Save the errors (unstandardized residual) and plot these against the dependent variable and eyeball for a funnel-like shape.

Errors are normally distributed
(optional)



No clear funnel
shape

Simple Scatter of commitment by Unstandardized Residual



Seems normally
distributed



Regression with dummy variables



Nominal independent variables

- DV of regression should be interval/ratio scaled. How about the IVs?
- Sometimes researchers are interested in categorical IV (e.g., sex, region, traveler status)
 - We can generate dummy (binary) variables coded as 0/1 and include the dummy variables as the independent variable.
 - However, when you include dummy variables, you must choose one category as the reference category.

Example:

We want to include gender (female/male) as a predictor in our regression model. However, the unit of gender has no meaning. Hence, we create a dummy variable **gender_dummy** in the following fashion:

- gender_dummy = 0 if gender = female;
- gender_dummy = 1 if gender = male.

Then we include gender_dummy into the regression model.

Regression with dummy variables

Gender

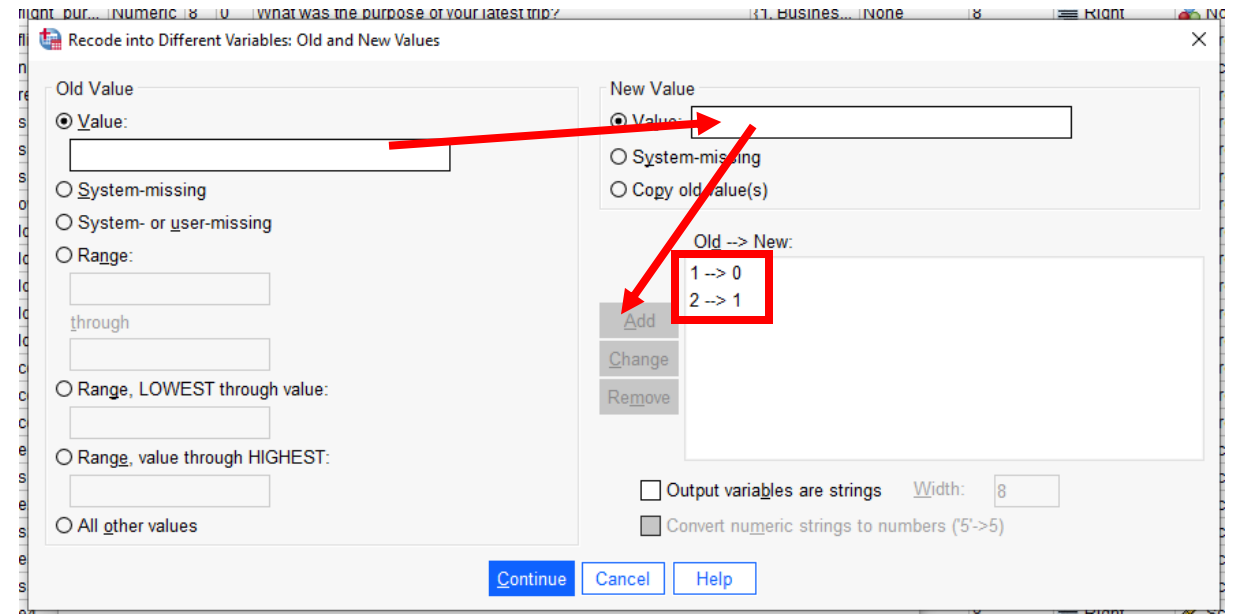
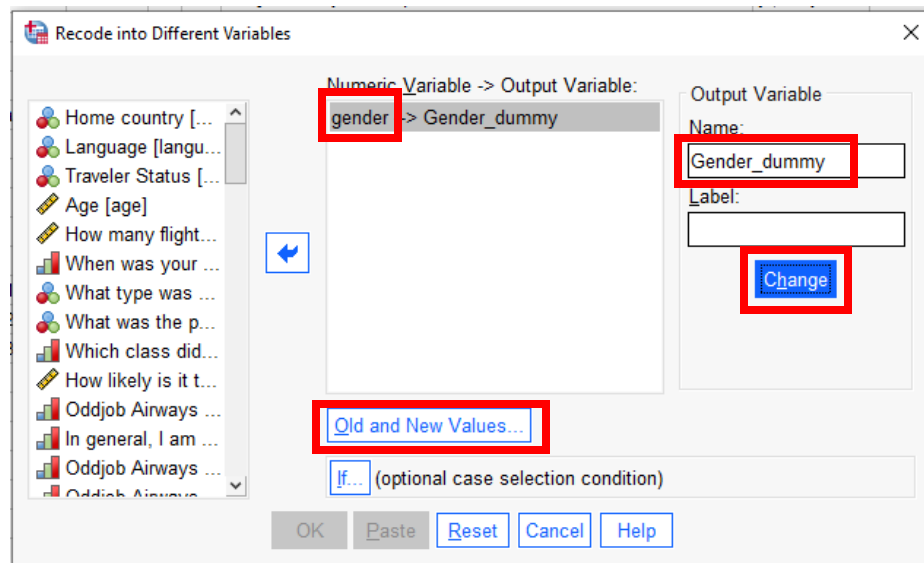
Value	Label
1	female
2	male



Gender_dummy

0 Female
1 Male

Transform → Recode into different variables → Select the variable of interest (gender) → Enter 'Gender_dummy' in the name → Click 'Change' → Click 'Old and New values'



Regression with dummy variables

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.570 ^a	.325	.321	1.41355

a. Predictors: (Constant), Gender_dummy, ... Oddjob Airways makes traveling uncomplicated., ... the condition of Oddjob Airways's aircraft is immaculate., ... Oddjob Airways offers great value for money., ... Oddjob Airways gives you a sense of safety., ... Oddjob Airways pays attention to its service delivery's details.

b. Dependent Variable: Commitment

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	929.922	6	154.987	77.566	<.001 ^b
	Residual	1930.192	966	1.998		
	Total	2860.114	972			

a. Dependent Variable: Commitment

b. Predictors: (Constant), Gender_dummy, ... Oddjob Airways makes traveling uncomplicated., ... the condition of Oddjob Airways's aircraft is immaculate., ... Oddjob Airways offers great value for money., ... Oddjob Airways gives you a sense of safety., ... Oddjob Airways pays attention to its service delivery's details.

• If user A is female, then her commitment = $21.324 + 23.588 * \text{looks_rating} + 0.137 * \text{days_active} - 3.480 * 0$

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	1.359	.189		7.187	<.001		
	... Oddjob Airways gives you a sense of safety. ^{s9}	.006	.003	.069	1.877	.061	.522	1.916
	... the condition of Oddjob Airways's aircraft is immaculate. ^{s10}	.001	.003	.017	.450	.653	.503	1.987
	... Oddjob Airways pays attention to its service delivery's details. ^{s19}	.012	.003	.156	4.131	<.001	.487	2.053
	... Oddjob Airways makes traveling uncomplicated. ^{s21}	.019	.003	.248	6.860	<.001	.533	1.876
	... Oddjob Airways offers great value for money. ^{s23}	.015	.003	.198	5.732	<.001	.587	1.702
	Gender_dummy	-.279	.103	-.072	-2.699	.007	.994	1.006

a. Dependent Variable: Commitment

Male travelers are *less* committed to Oddjob Airways by a difference of 0.279 compared to female travelers.



Regression model with dummy variable

- The coefficient of a **binary variable (0/1)** implies a shift of regression line
- If Traveler A is female, then her commitment = $1.359 + .012*s_{19} + .019*s_{21} + .015*s_{23} - .279*0$
- If Traveler B is male, then her commitment = $1.359 + .012*s_{19} + .019*s_{21} + .015*s_{23} - .279*1$
- If the only difference between traveler A and B is gender (A: female and B: male). Traveler A shows a higher commitment than Traveler B (on average).



Create (n-1) dummy variables for n-level variables

Status

Value Labels:

Value ▾	Label
1	Blue
2	Silver
3	Gold

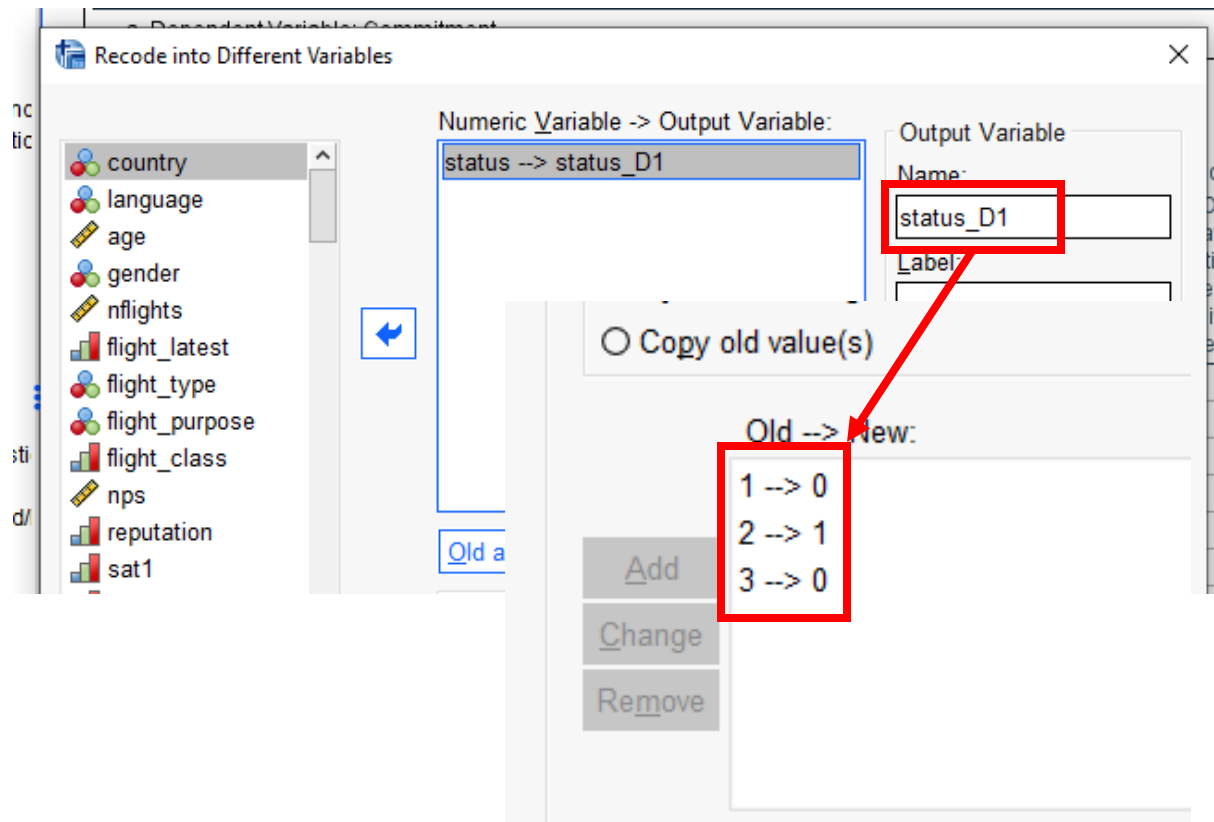
Levels	D1	D2
Blue (1)	0	0
Silver (2)	1	0
Gold (3)	0	1

$$Y = b_0 + b_1 D_1 + b_2 D_2$$

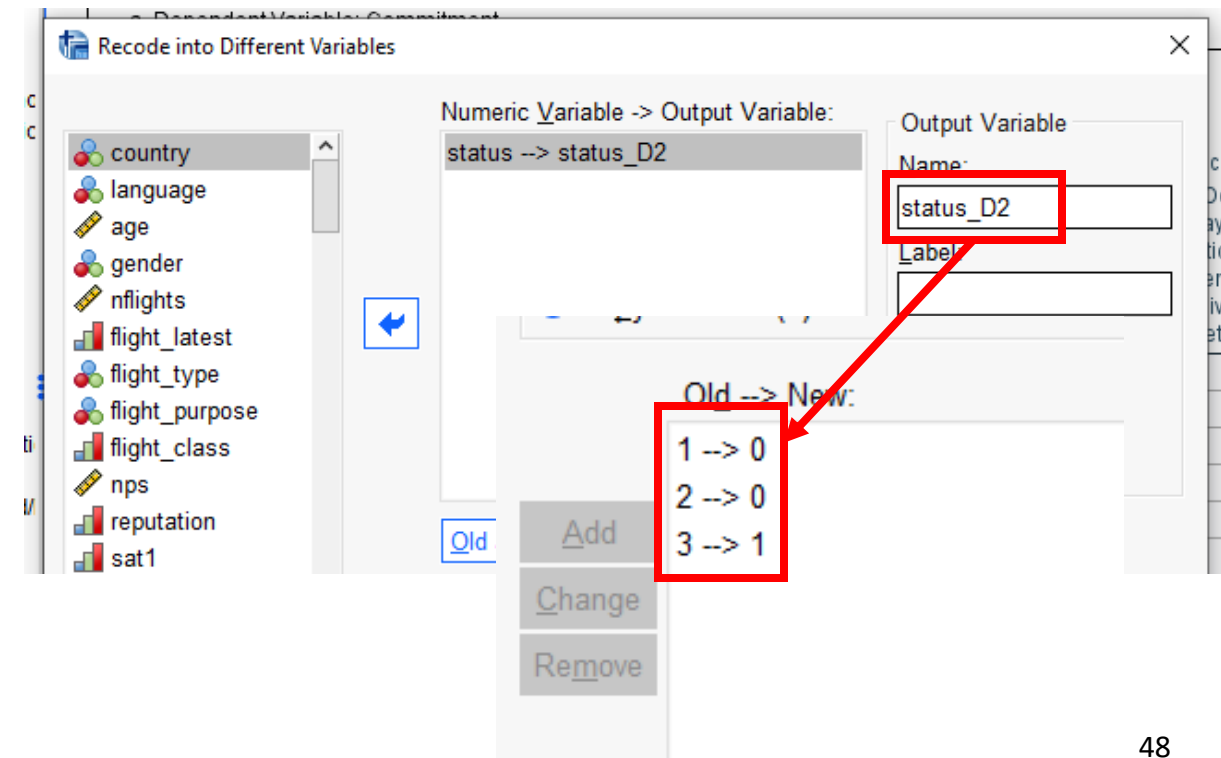
Regression with dummy variables

Transform → Recode into different variables → Select the variable of interest (Status) → Enter 'Status_D1' or 'Status_D2' in the name → Click 'Change' → Click 'Old and New values'

Status_D1



Status_D2



Regression with dummy variables

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.570 ^a	.325	.321	1.41394

a. Predictors: (Constant), status_D2, ... Oddjob Airways makes traveling uncomplicated., status_D1, ... the condition of Oddjob Airways's aircraft is immaculate., ... Oddjob Airways offers great value for money., ... Oddjob Airways gives you a sense of safety., ... Oddjob Airways pays attention to its service delivery's details.

b. Dependent Variable: Commitment

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	930.848	7	132.978	66.514	<.001 ^b
	Residual	1929.266	965	1.999		
	Total	2860.114	972			

a. Dependent Variable: Commitment

b. Predictors: (Constant), status_D2, ... Oddjob Airways makes traveling uncomplicated., status_D1, ... the condition of Oddjob Airways's aircraft is immaculate., ... Oddjob Airways offers great value for money., ... Oddjob Airways gives you a sense of safety., ... Oddjob Airways pays attention to its service delivery's details.

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	1.095	.180		6.098	<.001		
	... Oddjob Airways gives you a sense of safety. s9	.005	.003	.064	1.744	.081	.521	1.919
	... the condition of Oddjob Airways's aircraft is immaculate. s10	.001	.003	.012	.333	.739	.504	1.983
	... Oddjob Airways pays attention to its service delivery's details. s19	.012	.003	.147	3.867	<.001	.484	2.065
	... Oddjob Airways makes traveling uncomplicated. s21	.019	.003	.249	6.863	<.001	.532	1.879
	... Oddjob Airways offers great value for money. s23	.016	.003	.218	6.252	<.001	.574	1.741
	status_D1	.097	.111	.024	.880	.379	.936	1.068
	status_D2	.378	.137	.076	2.771	.006	.928	1.078

a. Dependent Variable: Commitment

Gold-status travelers are more committed to Oddjob Airways by a difference of 0.378 compared to Blue-status travelers.



Recap: Tests for Associations

Types of Test	Purpose of Test
Bivariate Regression Analysis	Analyze the linear relationship between <u>an</u> independent variable and <u>an interval/ratio</u> dependent variable.
Multiple Regression Analysis	Analyze the linear relationship between multiple independent variables and an interval/ratio dependent variable.



Workshop



Workshop Activity 1

1. Average the satisfaction items (sat1 ~ sat3)

*Transform – Compute Variable – Enter ‘**Satisfaction_avg**’ under Target Variable – Statistical – Mean (double click) - Select variables of interest*

2. Test the following hypothesis using a multiple regression

- H1: There is a linear association between satisfaction with on-board experience (s8) and overall satisfaction (Satisfaction_avg).
- H2: There is a linear association between satisfaction with exterior (s12) and overall satisfaction.
- H3: There is a linear association between satisfaction with hospitality (s17) and overall satisfaction.



Workshop activity 2

Test the following hypothesis using a *bivariate regression*

H1. Flight purpose has an impact on overall satisfaction (Satisfaction_avg)

* you may need to recode the IV

Transform → Recode into different variables → Select the variable of interest (flight_purpose) → Enter 'flight_purpose_dummy' in the name -> Click 'Change' -> Click 'Old and New values'

Is the hypothesis confirmed? (null rejected)



Workshop activity 3

Test the following hypothesis using a *regression*

H1. Flight class has an impact on overall satisfaction (Satisfaction_avg)

* you may need to recode the IV

Transform → Recode into different variables → Select the variable of interest (flight_class) → Enter 'flight_class_D1' or 'flight_class_D2' in the name -> Click 'Change' -> Click 'Old and New values'

Is the hypothesis confirmed? (null rejected)



Workshop activity 4

Present the result visually

H1. Gender has an impact on overall satisfaction (Satisfaction_avg)



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Thank you